Information Retrieval

Information retrieval finds useful items from a collection in response to a query. A document can be split into smaller passages called chunks. Chunking makes it easier to retrieve focused evidence, but chunks that are too small may lose context while chunks that are too large may contain unrelated information.

A retrieval system represents text and queries using features. TF-IDF gives more weight to terms that are frequent in a passage but less common across the collection. Cosine similarity compares vectors by their orientation and is commonly used to rank candidate passages.

Retrieval-Augmented Generation

Retrieval-augmented generation, usually abbreviated RAG, combines retrieval with a language model. The retriever selects passages relevant to a question. The generator uses those passages to formulate an answer. The retrieved passages are evidence, not a guarantee that the answer is correct.

A grounded RAG answer should be supported by the retrieved context and should cite its source. If evidence is missing or weak, the assistant should say that the documents do not contain enough information rather than inventing an answer.

Citations and Document Isolation

A citation should identify the document and page that support a claim. Page numbers should be retained during PDF extraction and chunking. If multiple documents are loaded, each passage should preserve its own document identity so the assistant does not attribute a statement to the wrong file.

Chat history can help resolve follow-up questions, but previous turns must not be treated as evidence for a new document-specific claim. Clear separation between conversation context and retrieved document evidence reduces source mixing.